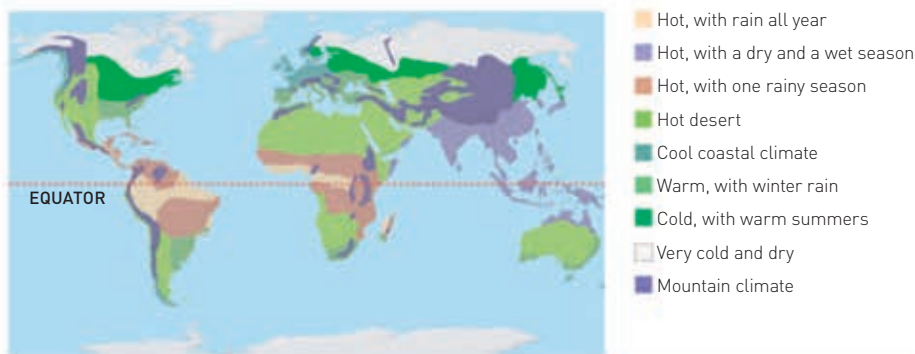


Climate and weather

Sunshine, air, and water interact to create the constantly changing conditions we call weather. Weather can change fast within a day and slowly from season to season. The average weather pattern in one place is what makes up its climate.

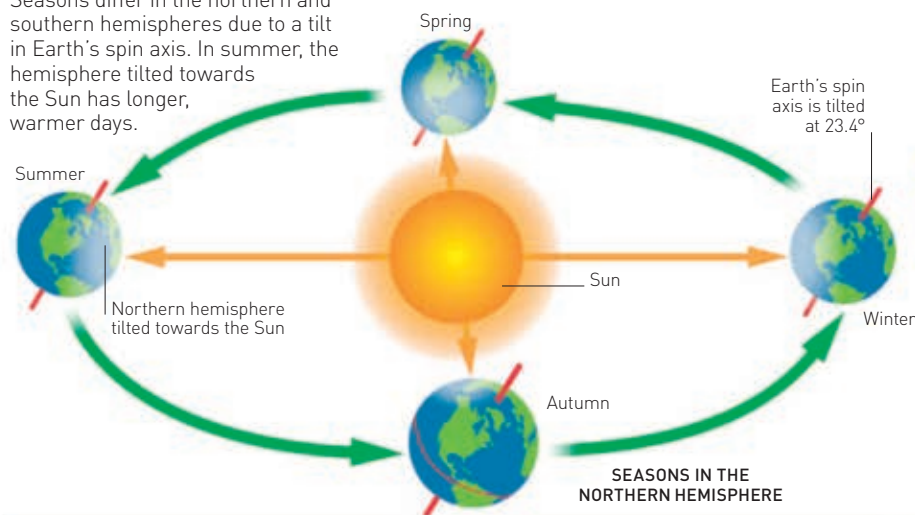
WORLD'S CLIMATE

The world is divided into climate zones, each one with a pattern of temperature and rainfall, and distinct vegetation. They range from a hot and wet climate near the equator to a cold and dry one at the poles.



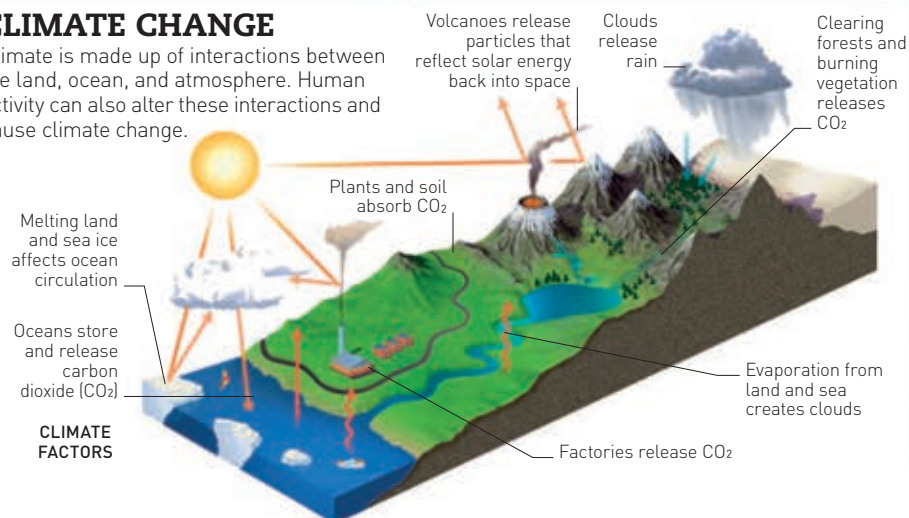
SEASONS

Seasons differ in the northern and southern hemispheres due to a tilt in Earth's spin axis. In summer, the hemisphere tilted towards the Sun has longer, warmer days.



CLIMATE CHANGE

Climate is made up of interactions between the land, ocean, and atmosphere. Human activity can also alter these interactions and cause climate change.



WHAT IS CLIMATE?

A climate is the average weather pattern in an area, influenced by factors such as the region's distance from the equator.



TEMPERATURE

Places tend to be colder the further they are from the equator.



PRECIPITATION

There are zones of high and low rainfall around the Earth.

WEATHER SYSTEMS

Local air masses have their own temperature, moisture content, density, and pressure. A weather front occurs when one air mass meets another.



WARM FRONT

Cold air is replaced by warm air, which slowly rises to form clouds and then rain as the air cools.



COLD FRONT

Cold air pushes into warm air, forcing it upwards to create storm clouds and heavy rain.

ATMOSPHERIC CIRCULATION

Air is always on the move, and this creates the circulation of the atmosphere around the globe. Hot air close to the equator is carried to high latitudes, and cool air is returned to the tropics. In each hemisphere, three "cells" with separate circulations create winds that blow from specific directions. These in turn produce surface currents in the oceans.

